DOI: 10.1111/eufm.12277

ORIGINAL ARTICLE

EUROPEAN FINANCIAL MANAGEMENT WILEY

Financial constraints and the growth and survival of innovative start-ups: An analysis of Italian firms

Edoardo Ferrucci[1] | **Roberto Guida**[2] | **Valentina Meliciani**[1]

[1]Department of Business and Management, University Luiss Guido Carli, Rome, Italy

[2]Department of Economics, University of International Studies, Rome, Italy

Correspondence
Roberto Guida, University of International Studies, via G. Baracconi 5A, 00161 Rome, Italy.
Email: r.guida@unint.eu

Abstract

We study the impact of measures devoted to relieving financial constraints for the growth and survival of Italian innovative start-ups. Using balance sheet data on innovative start-ups and information on the use of the Italian Central Guarantee Fund for small and medium-sized enterprises, we evaluate whether access to the fund, relieving financial constraints, helps innovative start-ups survive and grow. We find innovative start-ups benefit significantly more than similar control firms. We shed light on the relevance of policies aimed at reducing financial constraints for the growth and survival of innovative start-ups, an issue receiving increasing attention at the European level.

KEYWORDS

financial constraints, growth, innovative start-ups, survival

JEL CLASSIFICATION

*D*45; *G*14; *G*21; *G*32

We would like to thank an anonymous referee and the Editor (John Doukas) for the very helpful comments and suggestions on previous versions of the paper.

1 | INTRODUCTION

Innovative entrepreneurship is essential for sustained economic growth (Audretsch, Aldridge, & Sanders, 2011; Audretsch & Keilbach, 2004). From the viewpoint of evolutionary economics, entrepreneurs serve as agents of change, bring new ideas to markets, and stimulate growth through competitive firm selection (Audretsch, 1995; Jovanovic, 1982). However, these theories dismiss the importance of financial conditions in allowing entrepreneurship to act as a stimulus to economic growth, a point that Schumpeter (1912) made very clear in arguing that bank credit acts as money capital and, therefore, constitutes the necessary premise for the realization of the innovative processes planned by entrepreneurs.

While the financing of innovation plays a critical role in promoting economic growth, innovative firms often turn out to be financially constrained, because external finance can be very difficult or costly for them to obtain due to information asymmetry problems.[1] The argument goes as follows: innovative firms have better information about the probability of success and the nature of their innovative project than potential investors, so that external finance can be more expensive because it is harder for investors to distinguish good from bad projects (i.e., the lemon premium for innovation will be higher). Moreover, the revelation of information can represent a relevant cost to innovative firms because of the ease of imitation, which reduces the quality of their signal to the market (Bhattacharya & Ritter, 1983). In addition, innovative firms are characterized by a high share of intangible assets that cannot be pledged as collateral, and the investments in physical capital are often firm specific and have little collateral value (Hall, 2010). The information asymmetry problem can result in financial constraints, especially for innovative firms whose internal financial resources are limited, that is, small and young firms.

Based on the recognition of the importance of so-called young innovative companies (YICs) and their financial constraints, in 2016 the European Commission adopted an initiative to improve the economic and regulatory situation for young enterprises that have recently started or are in their early years of existence.[2]

With the same purpose, the Italian government, already in 2012, adopted focused measures favoring the birth and growth of YICs. In particular, Law No. 221/2012, the Italian Start-up Act, established a special section of the National Business Register to indicate the possibility for young innovative firms to be qualified as YICs (i.e., innovative start-ups). Among the various support measures that YIC registration provides, in addition to those related to tax incentives and exemptions, the 2012 law gives these firms priority and simplified access to a government-guaranteed bank loan fund – the Central Guarantee Fund (CGF) for small and medium-sized enterprises (SMEs) – to overcome market imperfections in the provision of bank loans to YICs. Such a specific guarantee facility allows 80% of the bank credit exposure of YICs to be covered, up to €2.5 million, and makes them fundable, independent of typical credit risk assessments, thanks to an overriding process.

But has this law actually helped YICs to overcome financial constraints? And has it increased firm life expectancy and growth rate? Despite the relevance of the policy for innovation and growth in Italy (one of the causes of Italy's productivity slowdown is its low rate of innovation), to our knowledge, the only studies evaluating the impact of the 2012 law policy

[1]For a review of the theoretical and empirical literature on the financing of innovation, see Hall (2010).

[2]See European Commission (2016), Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions – Europe's next leaders: the Start-up and Scale-up Initiative, Strasbourg, 22 November.

measure are those of Finaldi Russo, Magri, and Rampazzi (2016), Giraudo, Giudici, and Grilli (2019), and Biancalani, Czarnitzki, and Riccaboni (2020).

Based on propensity score matching, Finaldi Russo et al. (2016) find that external funding (either debt or equity) has increased more for YICs than for other similar firms since the 2012 law was enacted. Focusing on the effectiveness and potential interrelations of different policy instruments offered to YICs at the same time and in the same institutional context, Giraudo et al. (2019) highlight a sort of ‘institutional division of labor’ between the two main 2012 law facilities: firm access to a government-guaranteed bank loan and fiscal incentives for venture capital (VC) equity investments. Finally, Biancalani et al. (2020) find that YICs show a greater capacity to collect equity and debt than comparable untreated firms. However, none of these studies investigates the specific role of the CGF, and they neglect whether changed financial conditions affect firm growth and survival.[3]

By investigating the impact of the Italian Start-up Act (through access to the government-guaranteed bank loan fund) on firm growth and survival, this paper integrates the literature on financial constraints for innovative and risky firms (Hall, 2010; Hall & Lerner, 2010; Huyghebaert & Van de Gucht, 2011; Kerr, Lerner, & Schoar, 2011) with studies on the impact of financial constraints on growth and survival (e.g., Bottazzi, Secchi, & Tamagni, 2014; Cefis, Bartoloni, & Bonati, 2020; Liu & Li, 2017; Musso & Schiavo, 2008). In particular, it sheds light on whether relieving financial constraints is especially beneficial for the growth rate and survival of innovative start-ups with respect to non-innovative ones. The investigation of this issue, which has been neglected by the literature, has important policy implications, since it can inform policymakers on the opportunity to adopt special policies for YICs, a topic that has been receiving increasing attention at the European level (European Commission, 2016).

The empirical analysis is based on balance sheet data for innovative start-ups and for a control sample from the Bureau van Dijk's Aida database merged with information on the use of the CGF for Italian SMEs.

The remainder of the paper is organized as follows. Section 2 reviews the literature on financial constraints for young innovative firms and relates it to the literature on the growth and survival of small firms. Section 3 describes the dataset and provides descriptive statistics comparing the sample of innovative start-ups with the control group. Section 4 describes the methodology. Section 5 reports and comments on the results. Section 6 concludes the paper.

2 | LITERATURE REVIEW

Many strands of literature, both theoretical contributions and empirical evidence, are significant for the topic developed in this paper. The scientific debate is based on the presence of relevant capital market imperfections that particularly impact young and innovative firms (Hall, 2002; Revest & Sapio, 2012), lessening their capacity to access external resources and increasing their likelihood of ending up financially constrained. The literature investigates a causal link between financial constraints, investment in research and development (R&D), and innovation as well. The following reviews the most relevant works in these fields of research and highlights the main novelties of our contribution.

[3]A few studies in other countries have directly focused on the impact of support measures for start-ups, generally finding that these programs foster performance (Bertoni et al., 2019; Cantner & Kosters, 2015; Hottenrott & Richstein, 2020; Howell, 2017; Soderblom, Samuelsson, Wiklund, & Sandberg, 2015).

The literature suggests various reasons for a mismatch between the demand and supply of finance capital for a highly risky (i.e., young and innovative) firm that ends up financially constrained (Hall & Lerner, 2010). Among the determinants of the mismatch, studies highlight several factors, such as adverse selection and moral hazard problems, due to the presence of information asymmetries between the firm and potential investors (Brown, Fazzari, & Petersen, 2009; Carpenter & Petersen, 2002; Kerr et al., 2011; Leland & Pyle, 1977; Peneder, 2008), which can be caused by unstable cash flow or the lack of a solid track record or tangible assets (Hall, 2002; O'Sullivan, 2005). As found by Hall (2010), knowledge assets created by innovation investments are hard to use as collateral, since they are intangible, firm specific, and mainly embedded in human capital. Moreover, investors find it difficult to evaluate innovative projects' quality,[4] because banks have limited capacity to assess technologies in the early stages of adoption (Atanassov, Nanda, & Seru, 2007; Rajan & Zingales, 2003; Ueda, 2004).

Based on these theoretical considerations, most of the empirical literature uses investment cash flow sensitivity to test financial constraints on innovative firms (for a review, see Carreira & Silva, 2010). A number of studies find a significant positive cash flow effect on R&D investments, supporting the hypothesis that innovative firms are credit constrained. This is the case for US manufacturing firms (Hall, 1992) and small US firms in high-tech industries (Himmelberg & Petersen, 1994). Moreover, this result is not found for large firms (Hao & Jaffe, 1993). Brown et al. (2009) estimate dynamic R&D models for high-tech firms and find significant effects of cash flow and external equity for young, but not mature, firms. A positive cash flow effect on R&D investments is also found for French firms (Mulkay, Hall, & Mairesse, 2000) and Irish firms (Bougheas, Görg, & Strobl, 2003). Hillier, Pindado, de Queiroz, and de la Torre (2011) show that R&D at the firm level is less sensitive to internal cash flow in countries with effective investor protection, developed financial systems, and strong corporate control mechanisms. In the case of Italy, Scellato (2007) finds that only firms showing lower financial constraints, approximated by cash flow–investment sensitivity, are able to maintain a sustained patenting profile. In a longitudinal sample of 379 Italian unlisted new-technology-based firms (NTBFs), Bertoni, Colombo, and Croce (2010) observe that their investment rate is strongly positively correlated with their current cash flow and that, after receiving VC financing, they increase their investment rate. Colombo, Croce, and Guerini (2013) also find that the investment of small NTBFs (unlike that of large NTBFs) is sensitive to internal cash flows.

Financial constraints on innovation are also noted from survey data. The Community Innovation Survey carried out by European national statistical offices shows that the main obstacles to innovation in the majority of European firms are financial. In particular, the lack of appropriate sources of finance is one of the main obstacles to increasing both the probability and intensity of innovation throughout Europe (Canepa & Stoneman, 2004; D'Este, Iammarino, Savona, & von Tunzelmann, 2012; Mohnen, Palm, Loeff, & Tiwari, 2008; Mohnen & Roller, 2005; Savignac, 2009).

Given the importance, in the current paper, of the role of external funding sources for YICs, our work is also related to the literature on the role of the banking channel and other sources of financing for small firms (Berger & Udell, 1995; Petersen & Rajan, 1994, 1997). Along this line, Cosh, Cumming, and Hughes (2009) find that banks provide less capital to smaller risky ventures, while Freel (2007) finds that innovative firms are less successful in loan markets than their less innovative peers, confirming that they end up being financially constrained. Huyghebaert and Van de Gucht (2011) show that, in the context of business start-ups, banks finance a smaller fraction of debt when adverse

[4]The existence of market imperfections due to information problems implies that firms could be rationed by their lenders (Stiglitz & Weiss, 1981), resulting in a hierarchy (pecking order) of financial sources for firms (Myers & Majluf, 1984).

selection and risk-shifting incentives are potentially large, although these problems do not affect the maturity structure of bank loans. Ferrando, Marchica, and Mura (2017) show that financial flexibility attained through a conservative leverage policy is more important for investment for those companies that face higher external financing costs (small and young companies), while Bigelli, Martín-Ugedo, and Sánchez-Vidal (2014) find that financially conservative firms are smaller and have more intangible and fewer tangible assets. As for young innovative Italian firms, Magri (2009) suggests that supply factors can play a role in explaining the lower usage of bank loans, while Herrera and Minetti (2007) and Cosci, Meliciani, and Sabato (2016) find that strong relationships with banks promote innovation. On the importance of credit markets for the formation and success of young firms, Lemmon, Roberts, and Zender (2008) suggest that such firms are especially sensitive to changes in bank lending conditions, finding evidence of reliance on formal bank capital and its importance to firm growth. In the same strand of studies, Hirsch and Walz (2011) and Robb and Robinson (2012) show the importance of start-up external debt financing. Overall, although VC can be viewed as the most suitable form of external finance for YICs, bank loans remain the most important source of external finance for innovative firms in Europe (Colombo & Grilli, 2007; Giudici & Paleari, 2000).

Directly related to our paper are the studies by Finaldi Russo et al. (2016), Giraudo et al. (2019), and Biancalani et al. (2020) investigating the effects of the 2012 Italian law on YICs' financial structures. Using propensity score matching, Finaldi Russo et al. (2016) find that external funding (either debt or equity) has increased more for YICs than for other similar firms since the 2012 law. Focusing on the effectiveness and potential interrelations of different policy instruments offered to YICs at the same time and in the same institutional context, Giraudo et al. (2019) highlight a sort of institutional division of labor between the two main 2012 law facilities, namely, firm access to government-guaranteed bank loans and fiscal incentives for VC equity investments. Finally, Biancalani et al. (2020) find that YICs show greater capacity to collect equity and debt than comparable untreated firms. All these studies, however, neglect to investigate whether changed financial conditions affect overall firm performance or survival time.

Our paper asks whether relieving financial constraints truly helps innovative firms to grow and survive longer. Our contribution is therefore also related to studies investigating the problems of the growth and survival of YICs. The literature on firm growth finds financial constraints to have a negative impact on firm growth (Aghion, Fally, & Scarpetta, 2007; Bottazzi et al., 2014; Musso & Schiavo, 2008). Several studies also show that SMEs suffer more than large firms from financial constraints, which limits their growth rate (for a review, see Beck & Demirgüç-Kunt, 2006). But do innovative start-ups face more problems than other SMEs? While high-tech firms are more likely to grow (Morone & Testa, 2008; Westhead & Cowling, 1995), innovative activity is highly uncertain and, although it can increase the probability of superior performance, it cannot guarantee it (Coad & Rao, 2008). The risk associated with innovation activity can therefore limit firm growth, since innovative firms can be denied the financial resources necessary for their growth. Westhead and Storey (1997) find that, irrespective of the measure of technological sophistication chosen for a sample of independent high-tech firms, the most technologically sophisticated firms seemed much more likely to report that continual financial constraint impeded firm growth, compared with less technologically sophisticated firms. Hyytinena and Toivanen (2005) show that government funding in Finland is particularly beneficial for firms' R&D and growth in industries that depend more on external financing. The evidence is consistent with the view that financial constraint holds back innovation and growth.

Financial constraint can be detrimental not only for firms' growth, but also for their survival. The few studies including financial variables in equations modelling survival probabilities (Bridges & Guariglia, 2008; Bunn & Redwood, 2003; Cefis et al., 2020; Fotopoulos & Louri, 2000;

Liu & Lee, 2017; Musso & Schiavo, 2007; Vartia, 2004; Zingales, 1998) find a significant association between financial variables and survival in different countries. On the other hand, while the literature has found a robust positive relation between firms' survival and firms' size and age, the results on the role of innovation on survival are mixed. While most studies find a positive impact of innovation on firm survival (Cefis & Marsili, 2005, 2006; Esteve-Perez et al., 2007; Hall, 1987; Rosenbusch, Brinckmann, & Bausch, 2011), others find that the survival times of (especially small) innovative enterprises can be significantly shorter than those of non-innovative ones (Boyer & Blazy, 2014; Buddelmeyer, Jensen, & Webster, 2010; Cader & Leatherman, 2011; Hyytinena, Pajarinenb, & Rouvinen, 2015).

Linking the literature on the financial constraints of innovative firms with studies on the negative impact of financial constraint on firm growth and survival, we test whether relieving financial constraints is especially beneficial to the growth rate and survival probability of innovative start-ups.

3 | DATA AND DESCRIPTIVE STATISTICS

The empirical analysis is based on the following sources:

1. Comprehensive balance sheet data from Aida (the Italian version of the Bureau van Dijk's Amadeus database) and
2. A database on SMEs' access to the government-guaranteed bank loan facility, provided by the Italian Ministry of Economic Development.[5]

In 2012, the Italian Parliament approved a law aimed at stimulating the creation and development of innovative start-ups.[6] The law gives these firms priority and simplified access to a government-guaranteed bank loan fund (i.e., the CGF for SMEs) to overcome market imperfections in the provision of bank loans to YICs. Such a specific guarantee facility allows 80% of YIC's bank credit exposure to be covered, up to a limit of €2.5 million, and makes them fundable, independent of typical credit risk assessments, thanks to an overriding process. To gain access to this and other benefits, firms must meet eligibility criteria to register as innovative start-ups in a special section of the Business Register of the Chamber of Commerce. These criteria include being a limited liability company (either an Italian company or a branch of a European Union company registered in Italy), less than 5 years old, and operating in a technology-related business. In addition, YICs have to meet at least one of the following conditions: (a) the firm is a holder, depositary, or licensee of a patent or the owner and author of registered software; (b) at least one-third of the firm's employees are highly educated (with a PhD or a research tenure, or two-thirds have an MSc degree); and (c) the firm's annual R&D investments comprise at least 15% of its revenues (or operating costs, if higher).[7] Note that these firms can be easily identified, since they are registered in a special section of the Business Register of the Chamber of Commerce. Starting with an initial sample of 2,898 YICs, we downloaded their balance sheet data from Bureau van Dijk for the period from the year of their establishment to 2017.

[5]The ministerial decree for fast track access to the Central Guarantee Fund for SMEs was adopted in May 2013.

[6]See Law No. 221/2012.

[7]To be recognized as a YIC and maintain the right to belong to the Register, firms are prohibited from paying dividends or listing on a stock exchange. In addition, they cannot have revenues of over €5 million a year and must not have been created as a spin-off or a merger.

Since the analysis aims to study the effectiveness of the CGF in relieving financial constraints on YICs in comparison to other non-innovative start-ups (NISs), we also identify this second group of firms. To do so, we select a group of Italian limited liability companies less than 5 years old, with a production value of no more than €5 million (start-ups) but not registered as YICs in the special section of the Business Register of the Chamber of Commerce because they do not meet the criteria to be defined as an innovative start-up (i.e., operating in a technology-related business and meeting conditions (a), (b), or (c) above). We refer to this group of firms as NISs, and it consists of 15,789 companies. Since some firms might not strictly meet the criteria to be classified as YICs (or have not applied to be registered as such) but are nevertheless innovative, we perform robustness checks in the empirical analysis by deleting from the group of NISs all firms with positive R&D investments.

Table 1 breaks down the sample based on economic activity.[8]

As expected, YICs are highly concentrated in high-tech sectors: 36% operate in software development and information technology (IT) consultancy, while almost 20% are in R&D. The number of NISs that belong to these sectors is rather low (about 2% combined). On the other hand, the top sectors for NISs are trade, retail, and construction, where YICs are rather scarce (less than 5%).

In Table 2, we report the definitions of all the variables used in the empirical analysis and the set of financial and performance indicators used in the matching procedure.

In Table 3, we compare YICs and NISs based on all the variables used in the empirical analysis and the matching procedure, and in Table 4 we perform the same comparison for failed and non-failed firms.

We can see that YICs are younger and smaller and have a higher ratio of loans covered by the CGF and total assets (*CG Fund*), a higher ratio of intangible assets to total assets, and a higher growth rate with respect to NISs. Additionally, the profitability and financial structures of the two groups are extremely different: YICs are less profitable and more financially independent. These results hold across the samples of failed and non-failed firms. The main difference between the two samples is that the *CG Fund* ratio and the growth rate of YICs are larger than for NISs, but only among non-failed firms, whereas the opposite holds for failed firms. The high level of heterogeneity between the two samples is a concern for the empirical analysis and will be tackled through a matching procedure.

4 | ESTIMATED EQUATIONS AND METHODOLOGY

The empirical analysis is devoted to studying the impact of innovative start-ups' easy access to the CGF on their growth and survival rate. We therefore estimate two equations, one where the dependent variable is firm growth (Equation (1)) and the other where it is firm survival time (Equation (3)). In both equations, we estimate the impact of the ratio of loans covered by the CGF and total assets (*CG Fund*) to growth and survival and test whether it differs between YICs and NISs.

Several possible variables can be used to measure firm growth, but the most widely used in empirical studies are sales and employment (Delmar, 1997). While the consensus seems to be that, if only one indicator is chosen, the preferred measure of firm growth should be sales, sales does not always lead growth among high-tech start-ups, since the number of employees is more

[8]Firms are classified according to the 2007 ATECO version. For a complete description, see https://www.istat.it/it/archivio/17888

TABLE 1 Distribution of firms across sectors

This table reports the initial sample of firms by ATECO sector. The ATECO classification of sectors of economic activity was adopted by the Italian National Institute of Statistics in 2008, and it corresponds to NACE Rev. 2.

Activity description (ATECO, 2007)	Non-innovative	Non-innovative (%)	YIC	YIC (%)	All	All (%)
Wholesale trade	1,929	14.48%	31	1.24%	1,960	12.39%
Retail	1,651	12.39%	61	2.45%	1,712	10.82%
Buildings construction	1,363	10.23%	11	0.44%	1,374	8.69%
Software production and IT consultancy	250	1.88%	900	36.13%	1,150	7.27%
Specialized constructions	1,009	7.57%	31	1.24%	1,040	6.58%
Catering activities	960	7.20%	5	0.20%	965	6.10%
Land transport	655	4.92%	0	0.00%	655	4.14%
Research and development	36	0.27%	490	19.67%	526	3.33%
Repair of motor vehicles	523	3.92%	1	0.04%	524	3.31%
Manufacturing of metal products	504	3.78%	14	0.56%	518	3.27%
Real estate	505	3.79%	3	0.12%	508	3.21%
Information services	181	1.36%	241	9.67%	422	2.67%
Business management and consulting	337	2.53%	84	3.37%	421	2.66%
Other professional, scientific, and technical activities	262	1.97%	105	4.22%	367	2.32%
Support activities for office functions	297	2.23%	52	2.09%	349	2.21%
Manufacture of machinery and equipment	224	1.68%	101	4.05%	325	2.05%
Tests and technical analysis	207	1.55%	113	4.54%	320	2.02%
Storage and transport support activities	314	2.36%	4	0.16%	318	2.01%
Service activities for buildings and landscape	266	2.00%	3	0.12%	269	1.70%
Food industries	256	1.92%	13	0.52%	269	1.70%
Accommodation	249	1.87%	0	0.00%	249	1.57%
Supply of electricity, gas, steam	229	1.72%	20	0.80%	249	1.57%
Sports and entertainment activities	212	1.59%	3	0.12%	215	1.36%
Repair, maintenance, and installation of machinery	195	1.46%	14	0.56%	209	1.32%

(Continues)

TABLE 1 (Continued)

Activity description (ATECO, 2007)	Non-innovative	Non-innovative (%)	YIC	YIC (%)	All	All (%)
Health care	192	1.44%	5	0.20%	197	1.25%
Advertising and market research	153	1.15%	36	1.45%	189	1.19%
Manufacture of computers and electronic products	52	0.39%	127	5.10%	179	1.13%
Packaging of clothing articles	172	1.29%	3	0.12%	175	1.11%
Education	143	1.07%	20	0.80%	163	1.03%
Total	13,326		2,491		15,817	

likely to grow first in such firms (Delmar, Davidsson, & Gartner, 2003). Given the nature of the firms in our sample (i.e., YICs operating mainly in high-tech businesses, versus NISs operating in more traditional sectors), the empirical analysis on firm growth is repeated with different growth measures (sales growth and growth in the number of employees). We estimate the following equation:

$$\begin{aligned}\ln(Size)_{i,t} - \ln(Size)_{i,t-1} = \beta_0 + \beta_1 \ln(Size)_{i,t-1} + \beta_2 YIC_i + \beta_3 \ln(CG\ Fund)_{i,t-1} \\ + \beta_4 \ln(CG\ Fund)_{i,t-1}{}^{*}YIC_i + \beta_5 \ln(Age)_{i,t-1} + \beta_6 \ln(Age^2)_{i,t-1} \\ + \beta_7 \ln(Cash\ flow)_{i,t-1} + \beta_8 \ln(Intangibles)_{i,t-1} + e_{i,t-1}\end{aligned} \tag{1}$$

TABLE 2 Variable definitions

Name	Definition
YIC	Dichotomous variable that takes the value of one for innovative start-up firms, and zero otherwise
Age	Number of years since the firm's establishment year
Employees	Firm's total number of employees
Sales	Firm's value of revenues
CG Fund	Firm's ratio of loans covered by the CGF to total assets
Total assets	Firm's value of total assets
Earnings	Firm's value of total profits (or losses)
Intangibles	Firm's ratio of intangible assets to total assets
Cash flow	Firm's after-tax operating profits plus depreciation and amortization
Growth of sales	Annual growth rate of a firm's sales
ROA	Return on assets
Equity-to-asset ratio	Firm's ratio of equity to total assets
Debt ratio	Firm's ratio of total debt to equity
Operating income	Difference between a firm's gross income and operating expenses

TABLE 3 Descriptive statistics

This table reports descriptive statistics for all the variables used in the regression analysis and in the matching procedure. In the table, we perform a series of t-tests on the means between innovative start-ups (YIC) and non-innovative ones (non-YIC) and a series of nonparametric K-sample tests on the equality of the median values.
* denotes significance at the 0.01 level.

					Test on means			Test on medians		
Variables	**Obs.**	**Min.**	**Max.**	**Mean**	**Mean non-YIC**	**Mean YIC**	**Diff.**	**Median non-YIC**	**Median YIC**	**Chi-squared**
Age	140,729	0.000	10.000	4.488	4.500	4.386	0.114*	5.000	4.000	53.069*
Employees	112,801	0.000	1,518.000	5.998	6.421	2.265	4.155*	3.000	1.000	3,433.940*
Sales (thousands of euros)	115,215	0.000	166,151.000	673.771	728.574	191.973	536.601*	408.000	36.000	7,944.213*
CG Fund	117,109	0.000	13.342	0.203	0.175	0.442	−0.266*	0.000	0.000	280.5724*
Total assets (thousands of euros)	83,512	0.000	832,485.000	985.262	1,077.296	412.591	664.706*	427.000	108.000	4,296.617*
Earnings (thousands of euros)	115,211	−14,148.000	112,394.000	9.971	15.008	−34.304	49.311*	5.000	−1.000	3,066.023*
Intangibles	83,412	−0.024	1.000	0.086	0.057	0.269	−0.211*	0.007	0.178	5,722.334*
Cash flow (thousands of euros)	83,466	1.000	18,480.008	30.761	31.125	28.499	2.626	10.000	5.000	859.842*
Growth of sales	115,182	−6.672	8.392	0.325	0.321	0.374	−0.053*	0.121	0.288	258.034*
ROA	113,510	−956.010	603.560	2.867	4.777	−13.949	18.725*	4.140	−0.580	1,801.688*
Equity-to-asset ratio	115,074	−0.499	1.000	0.242	0.226	0.384	−0.157*	0.146	0.335	2,183.070*
Debt ratio	140,729	−1,988.660	9,790.000	14.704	15.491	7.790	7.701*	5.580	2.315	3,005.672*
Operating income (thousands of euros)	115,224	−7,063.000	88,836.000	27.590	34.140	−29.932	64.073*	15.000	0.000	4,767.977*

TABLE 4 *t*-Tests for failed versus non-failed firms

This table reports two sets of *t*-tests, depending on whether the firms belong to the group of failed firms or not. The table compares the means of the variables used in the regression analysis and in the matching procedure between innovative start-ups (YIC) and non-innovative ones (non-YIC). * denotes significance at the 0.01 level.

	Failed firms			Non-failed firms		
	Non-YIC	**YIC**	**Diff.**	**Non-YIC**	**YIC**	**Diff.**
Age	4.500	4.348	0.152*	4.500	4.388	0.112*
Employees	7.905	0.721	7.184*	5.984	2.152	3.832*
Sales	867.318	54.877	812.442*	682.168	177.808	504.359*
CG Fund	0.408	0.289	0.118*	0.142	0.468	−0.326*
Total assets	1,168.950	141.809	1,027.142*	1,022.323	387.230	635.093*
Earnings	0.760	−31.378	32.138	17.095	−32.687	49.781*
Intangibles	0.047	0.217	−0.170*	0.057	0.269	−0.211*
Cash flow	35.556	12.383	23.174*	29.159	26.725	2.434
Growth of sales	0.373	0.139	0.235*	0.330	0.385	−0.055*
ROA	0.839	−45.570	46.409*	5.441	−15.282	20.722*
Equity-to-asset ratio	0.182	0.448	−0.266*	0.232	0.379	−0.147*
Debt ratio	80.525	28.663	51.862	55.934	27.391	28.543*
Operating income	43.818	−84.596	128.415*	33.990	−16.666	50.656*
Percentage	15.370	11.890		84.630	88.110	

where ln indicates the natural logarithm of the variable; $Size_{i,t}$ is the value of sales (or the number of employees, for robustness checks) for firm i at time t; *YIC* is a dummy variable for innovative start-ups; *CG Fund* is the ratio of loans covered by the CGF to total assets; *Age* is the age of the firm at time $t-1$; *Cash flow* denotes funds collected within an enterprise from its operations, calculated as after-tax operating profits plus depreciation and amortization; *Intangibles* is the ratio of intangible assets to total assets; and ε is the error term. Since the dependent variable is the growth rate of the firm between times $t-1$ and t, all the explanatory variables are measured at time $t-1$. All specifications include sectoral, regional, juridical form, and calendar year dummies. Equation (1) is estimated with generalized least squares, with standard errors clustered at the firm level.

The coefficients of interest in this paper are the dummy for YICs (β_2), the coefficient on the logarithm of the *CG Fund* (β_3), and its interaction with the dummy for YICs (β_4). In particular, we have no strong a priori expectations for the coefficient β_2 of the YIC variable, since, on the one hand, innovative start-ups, when successful, can have good market opportunities and grow more than the control group, but, on the other hand, are potentially more credit constrained due to their risky activity, which can be detrimental for their growth rate. However, we expect that the greater intensity of CGF use is more beneficial for the growth of YICs than for NISs, since it helps relieve financial constraints that are expected to be more severe for the former. Thus, we expect a positive sign for the coefficient β_3.

As for the control variables, according to Gibrat's law, the firm's proportional rate of growth is independent of its initial size. If the law holds, the distribution of firm size is log-normal (with a large number of small firms and very few large firms). In that case, we should expect β_1 to equal zero. However, many empirical studies find that small firms tend to grow more than large ones (e.g., Santarelli, Klomp, & Thurik, 2006). The age of the firm is expected to negatively affect firms' growth (old firms are expected to grow less than young ones), although we do not rule out the possibility of nonlinear effects. The availability of internal funds (*Cash flow*) is expected to positively affect firm growth by providing means to expand sales (Bottazzi et al., 2014). A greater proportion of intangible assets can hamper growth when firms need to provide collateral to obtain credit (Bottazzi et al., 2014), but it can also have a positive effect, since intangibles are a major driver of firm competitiveness and are expected to sustain firms' long-run performance (Denicolai, Cotta Ramusino, & Sotti, 2015).

The second hypothesis we aim to test is whether the CGF helps innovative start-ups survive longer. In Aida, the survival status of firms is traceable until the end of 2017. For each firm, we construct a dichotomous variable $Fail_{i,t}$ that equals one if firm i is recorded as either dissolved, liquidated, or bankrupted in year t, and zero otherwise.

The standard estimation strategy to analyze firm survivability is the Cox proportional hazard model. This model relies on the definition of the survival function $S_i(t)$, which describes the probability of firm i surviving beyond a generic time period t^*, specifically $S_i(t) = Pr(T \geq t)$, where T is a continuous random variable that indicates the survival time and that assumes one of the values is in the range $[t_0, t^*]$. The hazard rate $H_i(t)$ describes the rate at which a firm fails in year t, given it was active in year $t-1$, and it is defined as $f_i(t)/S_i(t)$, where $f_i(t)$ is the probability density function of T. The hazard function in the Cox model has the form

$$H_i(t) = H_0(t)\exp(B'X_{it}) \tag{2}$$

where $H_0(t)$ is the baseline hazard and X is a list of covariates. While the model offers a high degree of flexibility, given that the parameters can be estimated without a specific functional form for the baseline hazard, $H_0(t)$, it also requires regressors in Equation (2) to be multiplicatively associated with hazard (i.e., proportional hazards assumption).

The multiplicity assumption is very stringent: given two random units from the sample, the ratio of hazards must be constant over time. We run two instances of Schoenfeld's test and in both cases we reject the null hypothesis of hazard rate proportionality.[9]

We therefore opt for a more conservative strategy through estimation of an accelerated failure time (AFT) model. Unlike proportional hazards models, the parameter estimates in AFT models are robust to omitted variable bias and, perhaps more importantly, are less affected by the choice of probability distribution (Keiding, Andersen, & Klein, 1997). The approach is frequently used in the literature on firm survival (e.g., Cader et al., 2011; Falck, 2007; Saridakis, Mole, & Storey, 2008; Strotmann, 2007).

In its log-linear form, our AFT model is given by

$$\begin{aligned}\log(T_i) = {} & \gamma_0 YIC_{i,} + \gamma_1 \ln(CG\ Fund)_{i,t-1} + \gamma_2 \ln(CG\ Fund)_{i,t-1}{}^*YIC_i + \gamma_3 \ln(Size)_{i,t-1} \\ & + \gamma_4 \ln(Age)_{i,t-1} + \gamma_5 \ln(Age^2)_{i,t-1} + \gamma_6 \ln(Growth)_{i,t-1} + \gamma_7 \ln(ROA)_{i,t-1} \\ & + \gamma_7 \ln(Equity\text{-}to\text{-}asset\ ratio)_{i,t-1} + \gamma_8 \ln(Debt\ ratio)_{i,t-1} + \alpha\varepsilon_i\end{aligned} \tag{3}$$

[9]The results are available on request.

where $\log(T_i)$ is the log-transformed survival time T while ε_i and α represent an error term and a scale parameter, respectively. To test whether greater intensity of use of the CGF affects the survival times of YICs and NISs differently, we include among the regressors a dummy for YICs, the ratio of loans covered by the CGF to total assets, and an interaction term between the two. Following the literature (e.g., Cefis & Marsili, 2005; Iwasaki & Kočenda, 2020), the richest specification includes as control variables firm size (measured by sales in the basic specification and by employees in robustness checks) and age (also allowing for nonlinear effects), firm growth (given the firm's size, the likelihood of survival could differ between firms growing and declining in size), and firm performance variables, such as the return on assets (ROA), the equity-to-asset ratio, and the debt ratio (ratio of total debt to assets).[10] All the variables are expressed in natural logarithms. In all estimations we also control for sectoral, regional, juridical form, and calendar year dummies.

5 | RESULTS

Table 5 reports the results for different specifications of Equation (1). Column (1) reports the results for a basic specification in which firm growth depends on firm size and age; the ratio of loans covered by the CGF and total assets (*CG Fund*); a dummy for YICs; the interaction term between the YIC dummy and *CG Fund*; and regional, juridical status, and sectoral dummies. Column (2) allows for the nonlinear effect of age on growth, and in column (3) we include controls for the intensity of intangible assets (*Intangibles*) and for a proxy of firms' ability to self-finance their growth (*Cash flow*).

Table 5 shows that small firms tend to grow more than large ones (size, measured by sales, always has a negative impact on growth, violating Gibrat's law). We also find that innovative start-ups grow, on average, less than similar start-ups (4.2–5.3 percentage points less), supporting the hypothesis that the most technologically sophisticated firms suffer more from financial constraints that limit their growth than less technologically sophisticated ones (Westhead & Storey, 1997). We find that growth decreases with age (in a specification, age has a non-monotonic effect on growth, with relatively younger and older firms growing more). Finally, we observe that the proportion of intangible assets and cash flow have a positive impact (Hall, 2002).

The results of all the specifications show that obtaining more funds from banks helps firms grow. The use of these funds helps innovative start-ups significantly more than non-innovative young firms, with an additional effect of around 0.025 (a 1% increase in the ratio of loans covered by the CGF and total assets increases firm growth by about 0.01% for NISs and by about 0.035% for YICs). This evidence provides indirect support for the hypothesis that, while SMEs suffer from financial constraints, which limit their growth rate (Beck & Demirgüç-Kunt, 2006), financial constraints are more severe for innovative firms (Hall, 2002; Revest & Sapio, 2012). Overall, access to a guarantee fund appears to be an effective way of overcoming information asymmetries favoring the growth of YICs.

However, does the CGF also help innovative start-ups live longer? Table 6 shows the results of the survival analysis (Equation (3)), where we report time ratios.[11] We can use time ratios to compare the rates at which different firms approach failure. Specifically, a time ratio greater than one for a predictor indicates that an increase in this predictor delays the firm's time to

[10]To apply a logarithmic transformation, we rescaled variables with negative values so that the minimum equals one.

[11]In all the specifications, we model survival times using a log-normal distribution. This is a standard choice in this literature (Che, Lu, & Tao, 2017). The choice of a log-logistic distribution provides very similar estimates.

TABLE 5 Regression results for the growth equation

This table reports the results of panel regressions of the growth equation specified in Equation (1). Explanatory variables are expressed in logarithms and are lagged by 1 year, and the coefficients are estimated with generalized least squares. All specifications include year, sectoral, regional, and firm juridical form dummies. The lower number of observations in column (3) is due to the missing values for the variables *Cash flow* and *Intangibles*. Robust standard errors are clustered at the firm level and are in parentheses. *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

	(1) Growth (sales)	(2) Growth (sales)	(3) Growth (sales)
Ln(*Sales*)	−0.334***	−0.333***	−0.268***
	(0.0037)	(0.0037)	(0.0051)
YIC	−0.535***	−0.519***	−0.421***
	(0.0231)	(0.0230)	(0.0211)
Ln(*CG Fund*)	0.012***	0.011***	0.007***
	(0.0008)	(0.0008)	(0.0007)
Ln(*CG Fund*) * *YIC*	0.023***	0.026***	0.0260***
	(0.0032)	(0.0032)	(0.0030)
Ln(*Age*)	−0.208***	−0.751***	−0.911***
	(0.0332)	(0.0655)	(0.0666)
Ln(Age^2)		0.286***	0.300***
		(0.0313)	(0.0306)
Ln(*Cash flow*)			0.044***
			(0.0026)
Ln(*Intangibles*)			0.162***
			(0.0255)
Intercept	2.700***	2.695***	2.233***
	(0.0287)	(0.0287)	(0.1080)
Year fixed effects (FE)	Yes	Yes	Yes
ATECO sector FE	Yes	Yes	Yes
Regional FE	Yes	Yes	Yes
Legal form FE	Yes	Yes	Yes
Observations	86,204	86,204	60,444
R^2	0.391	0.392	0.301
Log-likelihood	−87,285	−87,185	−54,640

failure (while the opposite applies for a time ratio lower than one). Table 6 reports different specifications: column (1) reports the results of the basic specification, in which the survival time depends on size (sales); age; sectoral, regional, time, and juridical form dummies; the ratio of loans covered by the CGF and total assets; a dummy for innovative start-ups; and its interaction with *CG Fund* (to investigate whether the impact of access to bank resources varies between innovative start-ups and similar young firms). In column (2), we allow for nonlinear effects of age and size and include a control for firm performance (i.e., firm growth). In column (3), we also control for a set of financial indicators (i.e., ROA, equity-to-asset ratio, and debt ratio).

A robust finding is that innovative start-ups have longer survival times with respect to non-innovative ones. The interpretation of the estimated time ratios is straightforward: innovative start-ups survive longer than non-innovative ones; specifically, those for which the dummy YIC equals one fail at a rate that is 50–70% slower. Again, the result is consistent with studies that identify innovation as one of the main drivers of firm survival (Cefis & Marsili, 2005, 2006; Esteve-Perez et al., 2007; Hall, 1987; Rosenbusch et al., 2011). As expected, survival increases with age, size, and firm growth. The ROA and the equity-to-asset ratio both have a small and positive effect on firm survival.

Focusing on the interaction term between *CG Fund* and the YIC dummy variable, we observe that *CG Fund* contributes to increasing firm survival only for YICs (for a unitary increase in the ratio of loans covered by CGF to total assets, the YIC failure rate decreases by about 3–5%). These results are partially in line with studies showing that financial variables matter for survival probabilities (Bridges & Guariglia, 2008; Bunn & Redwood, 2003; Fotopoulos & Louri, 2000; Musso & Schiavo, 2008; Vartia, 2004; Zingales, 1998). Overall, the result that the CGF contributes to increasing firm survival only for YICs (robust in all specifications) suggests an opportunity to establish specific policies directed at innovative start-ups to help them overcome market imperfections in the provision of bank loans and allow them not only to grow more, but also to survive longer.

The results reported in Tables 5 and 6 are based on a comparison between YICs and a group of non-innovative firms that are also start-ups (limited liability companies less than 5 years old, with a production value of no more than €5 million). However, as shown in section 3, YICs and NISs present serious heterogeneity, especially in terms of some financial indicators. To more accurately evaluate the impact of the CGF, we build a control sample of NISs that closely resembles the set of YICs.

We adopt the coarsened exact matching (CEM) algorithm described by Iacus, King, and Porro (2012). We select three financial indicators, namely, Operating income, Assets, and Earnings, plus a variable indicating the ATECO sector in which the firms operate. On the basis of these variables, we trim observations to reduce the imbalance between the YIC group and the group of NISs. We run the algorithm following the Freedman–Diaconis rule for distribution coarsening (Freedman & Diaconis, 1981), substantially reducing imbalance: the imbalance statistic described by Iacus et al. (2012) drops from an original value of 0.87 to 0.13. This strategy, although resulting in the exclusion of a portion of the data, given that the new matched sample consists of only 3,753 firms (1,917 NISs and 1,836 YICs), produces excellent results in terms of the overall balance of both financial indicators and the distribution of firms across ATECO sectors.[12]

In Tables 7 and 8, we replicate the econometric exercises described in Equations (1) and (3), respectively, over the matched samples.

The main results are confirmed: in the growth equation, innovative start-ups grow, on average, less than the control group; growth depends positively on the ratio of loans covered by the CGF and total assets for both YICs and the control group, but the advantage is more marked for YICs.

Looking at the estimates of the survival equation, we find that most of the results are confirmed. In agreement with previous estimations (see Table 6), borrowing increases the

[12]The results are available on request.

TABLE 6 Regression results for the survival equation

This table reports the results of panel regressions of the survival equation specified in Equation (3). Explanatory variables are expressed in logarithms and are lagged by 1 year, and we report time ratios estimated with an AFT model. All specifications include year, sectoral, regional, and firm juridical form dummies. Robust standard errors are clustered at the firm level, in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	(1) Survival	(2) Survival	(3) Survival
YIC	1.589***	1.720***	1.535***
	(0.0227)	(0.0128)	(0.0470)
Ln(*CG Fund*)	0.992	0.987	0.965
	(0.0030)	(0.0123)	(0.0222)
Ln(*CG Fund*) * *YIC*	1.194***	1.172***	1.215**
	(0.0711)	(0.0702)	(0.1020)
Ln(*Sales*)	1.052***	1.028***	1.038***
	(0.0055)	(0.0042)	(0.0046)
Ln(*Age*)	1.071***	1.069***	1.038**
	(0.0150)	(0.0162)	(0.0152)
Ln(Age^2)		1.000***	1.000***
		(0.0000)	(0.0000)
Growth of sales		1.347***	1.365***
		(0.0409)	(0.0441)
Ln(*ROA*)			1.006***
			(0.0008)
Ln(*Equity-to-asset ratio*)			1.007***
			(0.0011)
Ln(*Debt ratio*)			1.092
			(0.0626)
Year fixed effects (FE)	Yes	Yes	Yes
ATECO sector FE	Yes	Yes	Yes
Regional FE	Yes	Yes	Yes
Legal form FE	Yes	Yes	Yes
Observations	98,532	98,460	84,932
Firms	18,627	17,761	17,422
Failed	2,053	1,938	1,811
Chi-squared	40,124	37,494	36,426

probability of survival only for YICs. We emphasize once more the conclusion that favoring firms' access to bank loans through the CGF is particularly important for innovative start-ups suffering from financial constraint due to asymmetric information. At the same time, the results raise doubts about the advantages of the policy measure for the survival of SMEs in general.

TABLE 7 Regression results for the growth equation (matched sample)

This table reports the results of panel regressions of the growth equation specified in Equation (1). Explanatory variables are expressed in logarithms and are lagged by 1 year, and the coefficients are estimated with generalized least squares. All specifications include year, sectoral, regional, and firm juridical form dummies. The lower number of observations in column (3) is due to missing values for the variables *Cash flow* and *Intangibles*. The sample is obtained via CEM, as discussed in section 5. Robust standard errors are clustered at the firm level, in parentheses. *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

	(1) Growth (sales)	(2) Growth (sales)	(3) Growth (sales)
Ln(*Sales*)	−0.357***	−0.356***	−0.308***
	(0.0055)	(0.0055)	(0.0074)
YIC	−0.555***	−0.544***	−0.463***
	(0.0250)	(0.0249)	(0.0239)
Ln(*CG Fund*)	0.012***	0.0123***	0.008***
	(0.0012)	(0.0012)	(0.0012)
Ln(*CG Fund*) * *YIC*	0.022***	0.0251***	0.025***
	(0.0036)	(0.0036)	(0.0034)
Ln(*Age*)	−0.253***	−0.713***	−0.842***
	(0.0353)	(0.0718)	(0.0724)
Ln(Age^2)		0.239***	0.253***
		(0.0343)	(0.0336)
Ln(*Cash flow*)			0.043***
			(0.0039)
Ln(*Intangibles*)			0.218***
			(0.0374)
Intercept	2.956***	2.948***	2.604***
	(0.0440)	(0.0441)	(0.0649)
Year fixed effect (FE)	Yes	Yes	Yes
ATECO sector FE	Yes	Yes	Yes
Regional FE	Yes	Yes	Yes
Legal form FE	Yes	Yes	Yes
Observations	44,055	44,055	31,804
R^2	0.370	0.371	0.341
Log-likelihood	−41,265	−41,214	−31,986

We have also performed robustness checks for the growth and survival equations, respectively.[13] First, we measure size by the number of employees rather than sales; second, we exclude from the group of NISs (for both the whole sample and the matched sample) all firms

[13]The results of the robustness checks are available on request.

TABLE 8 Regression results for the survival equation (matched sample)

This table reports the results of panel regressions of the survival equation specified in Equation (3). Explanatory variables are expressed in logarithms and are lagged by 1 year, and we report time ratios estimated with an AFT model. All specifications include year, sectoral, regional, and firm juridical form dummies. The sample is obtained via CEM, as discussed in section 5. Robust standard errors are clustered at the firm level, in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	(1) Survival	(2) Survival	(3) Survival
YIC	1.188*** (0.0351)	1.241*** (0.0410)	1.326*** (0.0452)
Ln(*CG Fund*)	0.975 (0.0327)	0.953 (0.0357)	0.966 (0.0380)
Ln(*CG Fund*) * *YIC*	1.051*** (0.0085)	1.030*** (0.0085)	1.032*** (0.0090)
Ln(*Sales*)	1.042*** (0.0092)	1.026*** (0.0086)	1.025*** (0.0087)
Ln(*Age*)	1.100*** (0.0101)	1.112*** (0.0121)	1.124*** (0.0137)
Ln(Age^2)		1.000*** (0.0000)	1.000*** (0.0000)
Growth of sales		1.324*** (0.0183)	1.321*** (0.0188)
Ln(*ROA*)			1.003*** (0.0004)
Ln(*Equity-to-asset ratio*)			1.001** (0.0004)
Ln(*Debt ratio*)			1.000 (0.0001)
Year fixed effects (FE)	Yes	Yes	Yes
ATECO sector FE	Yes	Yes	Yes
Regional FE	Yes	Yes	Yes
Legal form FE	Yes	Yes	Yes
Observations	19,838	18,982	17,344
Firms	3,741	3,560	3,370
Failed	402	366	351
Chi-squared	26,514	23,661	22,240

with positive R&D investments. The main results of the paper are confirmed: use of the CGF increases growth and survival more for YICs than for NISs.

6 | CONCLUSIONS

This paper studies the impact of measures devoted to relieving financial constraints on the growth and survival of innovative start-ups in Italy. Taking advantage of the 2012 law allowing innovative start-ups that meet certain eligibility criteria to have simplified and prioritized access to the CGF for SMEs, we investigate whether use of the fund benefits these start-ups more than similar firms in the control group. While several studies have shown that SMEs suffer more than large firms from financial constraints, which limits their growth rate (for a review, see Beck & Demirgüç-Kunt, 2006), we provide indirect evidence that innovative start-ups face more problems than other SMEs. Although access to the CGF increases firm growth in general, its impact is much higher in the case of innovative start-ups. Moreover, the use of the fund reduces the hazard rate of innovative firms, but not that of similar SMEs.

Overall, this evidence supports the view that financial constraints are particularly severe for innovative firms due to the uncertainty that surrounds the innovation process, the high degree of information asymmetry, and the intangibility of the knowledge assets created by innovative investments. All these factors make it costly to carry out such investments using external finance. Although entrepreneurs could resort to internal sources, this can be difficult for start-ups and could limit their growth potential. Moreover, in Italy, as in many other bank-based continental European countries, the capital market is not yet a valid alternative to bank loans, and VC represents a small share of firms' funds. Given the importance of innovation for long-term growth and the financial problems faced by innovative firms, this is an area where policymakers at the national and European levels can introduce effective measures to mitigate financial constraints and foster the survival and growth of innovative firms.

Although we have shown that the use of the CGF has helped innovative start-ups grow and survive longer, further studies could evaluate the costs and benefits of different policies devoted to alleviating financial constraints, including those favoring the financing of innovative start-ups in the financial market and the growth of VC. Moreover, given the European Commission's interest in improving the economic and regulatory situation for young enterprises and the initiatives adopted in many European countries, similar analyses could be performed in other economies to determine the extent to which the results for Italy can be generalized to other bank- and market-based systems.

REFERENCES

Aghion, P., Fally, T., & Scarpetta, S. (2007). Credit constraints as a barrier to the entry and post-entry growth of firms. *Economic Policy*, *22*(52), 732–779.

Atanassov, J., Nanda, V., & Seru, A. (2007). *Finance and innovation: The case of publicly traded firms* (Ross School of Business Paper No. 970).

Audretsch, D. B. (1995). *Innovation and industry evolution*. Cambridge, MA: MIT Press.

Audretsch, D. B., Aldridge, T., & Sanders, M. (2011). Social capital building and new business formation: A case study in Silicon Valley. *International Small Business Journal: Researching Entrepreneurship*, *29*(2), 152–169.

Audretsch, D. B., & Keilbach, M. (2004). Entrepreneurship capital and economic performance. *Regional Studies*, *38*, 949–959.

Beck, T., & Demirgüç-Kunt, A. (2006). Small and medium-size enterprises: Access to finance as a growth constraint. *Journal of Banking & Finance*, *30*(11), 2931–2943.

Berger, N. A., & Udell, G. F. (1995). Small firms, commercial lines of credit, and collateral. *Journal of Business*, *68*(3), 351–382.

Bertoni, F., Jose, M., & Carmelo, R. (2019). The impact of government-supported participative loans on the growth of entrepreneurial ventures. *Research Policy*, *48*(1), 371–384.

Bertoni, F., Colombo, M. G., & Croce, A. (2010). The effect of venture capital financing on the sensitivity to cash flow of firm's investments. *European Financial Management*, *16*(4), 528–551.

Bhattacharya, S., & Ritter, R. (1983). Innovation and communication: Signalling with partial disclosure. *The Review of Economic Studies*, *50*(2), 331–346.

Biancalani, F., Czarnitzki, D., & Riccaboni, M. (2020). *The Italian Startup Act: Empirical evidence of policy effects* (Leuven Working Paper 648452). Department of Management, Strategy and Innovation.

Bigelli, M., Martín-Ugedo, J. F., & Sánchez-Vidal, F. J. (2014). Financial conservatism of private firms. *Journal of Business Research*, *67*, 2419–2427.

Bottazzi, G., Secchi, A., & Tamagni, F. (2014). Financial constraints and firm dynamics. *Small Business Economics*, *42*, 99–116.

Bougheas, S., Görg, H., & Strobl, E. (2003). Is R&D financially constrained? Theory and evidence from Irish manufacturing. *Review of Industrial Organization*, *22*(2), 159–174.

Boyer, T., & Blazy, R. (2014). Born to be alive? The survival of innovative and non-innovative French micro-start-ups. *Small Business Economics*, *42*(4), 669–683.

Bridges, S., & Guariglia, A. (2008). Financial constraints, global engagement, and firm survival in the U.K.: Evidence from micro data. *Scottish Journal of Political Economy*, *55*, 444–464.

Brown, J. R., Fazzari, S. M., & Petersen, B. C. (2009). Financing innovation and growth: Cash flow, external equity, and the 1990s R&D boom. *The Journal of Finance*, *64*(1), 151–185.

Buddelmeyer, H., Jensen, P. H., & Webster, E. (2010). Innovation and the determinants of company survival. *Oxford Economic Papers*, *62*(2), 261–285.

Bunn, P., Redwood, & V. (2003). *Company accounts based modelling of business failures and the implications for financial stability* (Bank of England Working Paper 210).

Cader, H. A., & Leatherman, J. C. (2011). Small business survival and sample selection bias. *Small Business Economics*, *37*(2), 155–165.

Canepa, A., & Stoneman, P. (2004). Financing constraints in the inter firm diffusion of new process technologies. *The Journal of Technology Transfer*, *30*(1–2), 159–169.

Cantner, U., & Kosters, S. (2015). Public R&D support for newly founded firms: Effects on patent activity and employment growth. *Journal of Innovation Economics & Management*, *1*, 7–37.

Carpenter, R. E., & Petersen, B. C. (2002). Capital market imperfections, high-tech investment, and new equity financing. *The Economic Journal*, *112*(477), F54–F72.

Carreira, C., & Silva, F. (2010). No deep pockets: Some stylized empirical results on firms' financial constraints. *Journal of Economic Surveys*, *24*(4), 731–753.

Cefis, E., Bartoloni, E., & Bonati, M. (2020). *Show me how to live*: Firms' financial conditions and innovation during the crisis. *Structural Change and Economic Dynamics*, *52*, 63–81.

Cefis, E., & Marsili, O. (2005). A matter of life and death: Innovation and firm survival. *Industrial and Corporate Change*, *14*(6), 1167–1192.

Cefis, E., & Marsili, O. (2006). Survivor: The role of innovation in firms' survival. *Research Policy*, *35*(5), 626–641.

Che, Y., Lu, Y., & Tao, Z. (2017). Institutional quality and new firm survival. *Economics of Transition*, *25*(3), 495–525.

Coad, A., & Rao, R. (2008). Innovation and firm growth in high-tech sectors: A quantile regression approach. *Research Policy*, *37*(4), 633–648.

Colombo, M. G., Croce, A., & Guerini, M. (2013). The effect of public subsidies on firms' investment–cash flow sensitivity: Transient or persistent? *Research Policy*, *42*, 1605–1623.

Colombo, M. G., & Grilli, L. (2007). Funding gaps? Access to bank loans by high-tech startups. *Small Business Economics*, *29*(1–2), 25–46.

Cosci, S., Meliciani, V., & Sabato, V. (2016). Relationship lending and innovation: Empirical evidence on a sample of European firms. *Economics of Innovation and New Technology*, *25*(3), 335–357.

Cosh, A., Cumming, D., & Hughes, A. (2009). Outside entrepreneurial capital. *Economic Journal*, *119*(540), 1494–1533.

Delmar, F. (1997). Measuring growth: Methodological considerations and empirical results. In R. Donckels & A. Miettinen (Eds.), *Entrepreneurship and SME research: On its way to the next millennium* (pp. 199–216). Brooksfield, Vermont: Ashgate Publishing Company.

Delmar, F., Davidsson, P., & Gartner, W. B. (2003). Arriving at the high-growth firm. *Journal of Business Venturing*, *18*, 189–216.

Denicolai, S., Cotta Ramusino, E., & Sotti, F. (2015). The impact of intangibles on firm growth. *Technology Analysis & Strategic Management*, *27*(2), 219–236.

D'Este, P., Iammarino, S., Savona, M., & von Tunzelmann, N. (2012). What hampers innovation? Revealed barriers versus deterring barriers. *Research Policy*, *41*(2), 482–488.

EU Commission. (2016). Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions–A European Agenda for the collaborative economy. Brussel 2, 2016.

Falck, O. (2007). Survival chances of new businesses: Do regional conditions matter? *Applied Economics*, *39*(16), 2039–2048.

Ferrando, A., Marchica, M. T., & Mura, R. (2017). Financial flexibility and investment ability across the Euro Area and the UK. *European Financial Management*, *23*(1), 87–126.

Finaldi Russo, P., Magri, S., & Rampazzi, C. (2016). *Innovative start-ups in Italy: Their special features and the effects of the 2012 law* (Banca d'Italia Occasional Papers 339).

Fotopoulos, G., & Louri, H. (2000). Location and survival of new entry. *Small Business Economics*, *14*(4), 311–321.

Freedman, D., & Diaconis, P. (1981). On the histogram as a density estimator: L_2 theory. *Probability Theory and Related Fields*, *57*(4), 453–476.

Freel, M. S. (2007). Are small innovators credit rationed? *Small Business Economics*, *28*(1), 23–35.

Giraudo, E., Giudici, G., & Grilli, L. (2019). Entrepreneurship policy and the financing of young innovative companies: Evidence from the Italian Startup Act. *Research Policy*, *48*, 103801.

Giudici, G., & Paleari, S. (2000). The provision of finance to innovation: A survey conducted among Italian technology-based small firms. *Small Business Economics*, *14*(1), 37–53.

Hall, B. H. (1987). The relationship between firm size and firm growth in the US manufacturing sector. *The Journal of Industrial Economics*, *35*(4), 583–606.

Hall, B. H. (1992). *Research and development at the firm level: Does the source of financing matter?* (NBER Working Paper 4096).

Hall, B. H. (2002). The financing of research and development. *Oxford Review of Economic Policy*, *18*(1), 35–51.

Hall, B. H. (2010). The financing of innovative firms. *Review of Economics and Institutions*, *1*(1), 1–30.

Hall, B. H., & Lerner, J. (2010). The financing of R&D and innovation. In B. Hall & N. Rosenberg (Eds.), *Handbook of the economics of innovation* (1, pp. 609–639). Amsterdam, Netherlands: Elsevier-North Holland.

Hao, K. Y., & Jaffe, A. B. (1993). Effect of liquidity on firms' R&D spending. *Economics of Innovation and New Technology*, *2*(4), 275–282.

Herrera, A. M., & Minetti, R. (2007). Informed finance and technological change: Evidence from credit relationships. *Journal of Financial Economics*, *83*(1), 223–269.

Hillier, D., Pindado, J., de Queiroz, V., & de la Torre, C. (2011). The impact of country-level corporate governance on research and development. *Journal of International Business Studies*, *42*, 76–98.

Himmelberg, C. P., & Petersen, B. C. (1994). R&D and internal finance: A panel study of small firms in high-tech industries. *Review of Economics and Statistics*, *76*(1), 38–51.

Hirsch, J., & Walz, U. (2011). Financing decisions along a firm's life-cycle: Debt as a commitment device. *European Financial Management*, *17*(5), 898–927.

Hottenrott, H., & Richstein, R. (2020). Start-up subsidies: Does the policy instrument matter? *Research Policy*, *49*, 103888.

Howell, S. T. (2017). Financing innovation: Evidence from R&D grants. *American Economic Review*, *107*, 1136–1164.

Huyghebaert, N., & Van de Gucht, L. M. (2011). The determinants of financial structure: New insights from business start-ups. *European Financial Management*, *13*(1), 101–133.

Hyytinena, A., Pajarinenb, M., & Rouvinen, P. (2015). Does innovativeness reduce startup survival rates? *Journal of Business Venturing*, *30*(4), 564–581.

Hyytinena, A., & Toivanen, O. (2005). Do financial constraints hold back innovation and growth?: Evidence on the role of public policy. *Research Policy*, *34*(9), 1385–1403.
Iacus, S. M., King, G., & Porro, G. (2012). Causal inference without balance checking: Coarsened exact matching. *Political Analysis*, *20*(1), 1–24.
Iwasaki, I., & Kočenda, E. (2020). Survival of service firms in European emerging economies. *Applied Economics Letters*, *27*(4), 340–348.
Jovanovic, B. (1982). Selection and the evolution of industry. *Econometrica*, *50*(3), 649–670.
Keiding, N., Andersen, P. K., & Klein, J. P. (1997). The role of frailty models and accelerated failure time models in describing heterogeneity due to omitted covariates. *Statistics in Medicine*, *16*(2), 215–224.
Kerr, W. R., Lerner, J., & Schoar, A. (2011). The consequences of entrepreneurial finance: Evidence from angel financings. *The Review of Financial Studies*, *27*(1), 20–55.
Leland, H. E., & Pyle, D. H. (1977). Informational asymmetries, financial structure, and financial intermediation. *The Journal of Finance*, *32*(2), 371–387.
Lemmon, M. L., Roberts, M. R., & Zender, J. F. (2008). Back to the beginning: Persistence and the cross-section of corporate capital structure. *The Journal of Finance*, *63*(4), 1575–1608.
Liu, X., & Li, H. (2017). Financial constraints and the productivity–survival link: Evidence from China's firm-level data. *Industrial and Corporate Change*, *26*(5), 763–779.
Magri, S. (2009). The financing of small innovative firms: The Italian case. *Economics of Innovation and New Technology*, *18*(2), 181–204.
Mohnen, P., Palm, F., Loeff, S., & Tiwari, A. (2008). Financial constraints and other obstacles: Are they a threat to innovation activity? *De Economist*, *156*(2), 201–214.
Mohnen, P., & Roller, L. H. (2005). Complementarities in innovation policy. *European Economic Review*, *49*(6), 1431–1450.
Morone, P., & Testa, G. (2008). Firms growth, size and innovation an investigation into the Italian manufacturing sector. *Economics of Innovation and New Technology*, *17*(4), 311–329.
Mulkay, B., Hall, B. H., & Mairesse, J. (2000). *Firm level investment and R&D in France and the United States: A comparison* (NBER Working Paper 8038).
Musso, P., & Schiavo, S. (2008). The impact of financial constraints on firm survival and growth. *Journal of Evolutionary Economics*, *18*(2), 135–149.
Myers, S. C., & Nicholas, S. M. (1984). Corporate financing and investment decisions when firms have information that investors do not have. No. w1396. National Bureau of Economic Research.
O'Sullivan, M. (2005). Finance and innovation. In J. Fagerberg, D. Mowery & R. Nelson (Eds.), *The Oxford handbook of innovation* (pp. 240–265). Oxford, UK: Oxford University Press.
Peneder, M. (2008). The problem of private under-investment in innovation: A policy mind map. *Technovation*, *28*(8), 518–530.
Petersen, M. A., & Rajan, R. G. (1994). The benefits of lending relationships: Evidence from small business data. *The Journal of Finance*, *49*(1), 3–37.
Petersen, M. A., & Rajan, R. G. (1997). Trade credit: Theories and evidence. *The Review of Financial Studies*, *10*(3), 661–691.
Rajan, R. G., & Zingales, L. (2003). The great reversals: The politics of financial development in the twentieth century. *Journal of Financial Economics*, *69*(1), 5–50.
Revest, V., & Sapio, A. (2012). Financing technology-based small firms in Europe: What do we know? *Small Business Economics*, *39*(1), 179–205.
Robb, A. M., & Robinson, D. T. (2012). The capital structure decisions of startup firms. *The Review of Financial Studies*, *1*(1), 1–27.
Rosenbusch, N., Brinckmann, J., & Bausch, A. (2011). Is innovation always beneficial? A meta-analysis of the relationship between innovation and performance in SMEs. *Journal of Business Venturing*, *26*(4), 441–457.
Santarelli, E., Klomp, L., & Thurik, A. R. (2006). Gibrat's law: An overview of the empirical literature. In E. Santarelli (Ed.), *Entrepreneurship, growth, and innovation. International studies in entrepreneurship* (12, pp. 41–73). Boston, MA: Springer.
Saridakis, G., Mole, K., & Storey, D. J. (2008). New small firm survival in England. *Empirica*, *35*(1), 25–39.
Savignac, F. (2009). Impact of financial constraints on innovation: What can be learned from a direct measure? *Economics of Innovation and New Technology*, *17*(6), 553–569.

Scellato, G. (2007). Patents, firm size and financial constraints: An empirical analysis for a panel of Italian manufacturing firms. *Cambridge Journal of Economics*, *31*(1), 55–76.

Schumpeter, J. (1912). *Theory of economic development*. New Brunswick, NJ: Transaction.

Soderblom, A., Samuelsson, M., Wiklund, J., & Sandberg, R. (2015). Inside the black box of outcome additionality: Effects of early-stage government subsidies on resource accumulation and new venture performance. *Research Policy*, *44*, 1501–1512.

Stiglitz, J. E., Andrew, W. (1981). Credit rationing in markets with imperfect information, Part I. American economic review.

Strotmann, H. (2007). Entrepreneurial survival. *Small Business Economics*, *28*(1), 87–104.

Ueda, M. (2004). Banks versus venture capital: Project evaluation, screening, and expropriation. *The Journal of Finance*, *59*(2), 601–621.

Vartia, L. (2004). *Assessing plant entry and exit dynamics and survival – Does firms' financial status matter?* (Mimeograph). European University Institute.

Westhead, P., & Cowling, M. (1995). Employment change in independent owner-managed high-technology firms in Great Britain. *Small Business Economics*, *7*(2), 111–140.

Westhead, P., & Storey, D. J. (1997). Financial constraints on the growth of high technology small firms in the United Kingdom. *Applied Financial Economics*, *7*(2), 197–201.

Zingales, L. (1998). Survival of the fittest or the fattest? Exit and financing in the trucking industry. *The Journal of Finance*, *53*(3), 905–938.

How to cite this article: Ferrucci E, Guida R, Meliciani V. Financial constraints and the growth and survival of innovative start-ups: An analysis of Italian firms. *Eur Financ Manag*. 2021;27:364–386. https://doi.org/10.1111/eufm.12277